

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations

May – June 2025

22MCA222 - HEALTHCARE ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer any Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Summarize the need and Career Scope in Healthcare Analytics with the real-time examples.	10	L*2	1	1
b)	Define the concept of data and information in healthcare analytics. Describe the key measurements, metrics, and indicators used in healthcare analytics.	10	L1	1	1
2. a)	Compare descriptive, predictive, and prescriptive analytics techniques in healthcare with respect to the data analysis methods and application areas. Explain each with an example.	10	L2	1	1
b)	Explain the role and significance of the Digital Imaging and Communications in Medicine (DICOM) standard in healthcare analytics.	10	L1	1	1
Unit – II					
3. a)	Define cross-correlation analysis, and explain the key steps of pre-Processing in cross-correlation analysis.	10	L2	2	1
b)	Explain the key concepts and characteristics of bio signals in the time domain and frequency domain in healthcare analytics.	10	L2	2	1
4. a)	List and explain the types of Biomedical Signals along with the devices and their applications.	10	L2	2	1
b)	What are the five significant challenges faced in implementing analytics in the healthcare industry, and how do they impact in healthcare decision-making?	10	L1	2	1
Unit – III					
5. a)	Explain the significance of genomic data analysis in personalized medicine. Discuss how genomic data can be used in treatment decisions and improve patient outcomes.	10	L2	3	2

P.T.O.

b)	Explain the different Stages of data integration, highlighting the advantages and their disadvantages.	10	L2	3	2
6. a)	Explain Clinicogenomic Integration? Discuss the issues and challenges in Integrating Clinical and Genomic data.	10	L2	3	2
b)	Outline an overview of the steps involved in the analysis of genomic data for personalized medicine. Explain the data preprocessing, feature selection, and modeling techniques used in genomic data analytics.	10	L2	3	2

Unit – IV

7. a)	Explain the concept of Natural Language Processing (NLP) in the context of biomedical information mining. Describe the core NLP components.	10	L2	4	2
b)	What are Electronic Health Records (EHRs)? Discuss the challenges and benefits of EHR.	10	L2	4	2
8. a)	Discuss the role of social media for the analysis, detection and tracking of Infectious disease outbreaks.	10	L2	4	2
b)	Discuss the social media analysis for public health research.	10	L2	4	2

Unit – V

9. a)	What are Alternative clinical prediction models, and how they are used in healthcare data analytics? Discuss their advantages and limitations.	10	L2	5	2
b)	Explain the concept of statistical prediction models in biomedical advanced healthcare data analytics. Discuss their significance in clinical decision-making and the challenges associated with their implementation.	10	L2	5	2
10. a)	Discuss the importance of visual analytics in healthcare and its applications.	10	L2	5	2
b)	What is temporal data mining, and how is it used in healthcare data analytics? Explain the temporal pattern mining techniques in detail.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
